

HDOC DAY 36

Problem

Write a menu-driven C program to perform the following operations using separate functions:

1. Count characters in a string without using built-in length functions.
2. Print each character of a string on a new line.
3. Count vowels and consonants in a string.
4. Exit.

C code

```
#include <stdio.h>

#include <ctype.h>

#include <string.h>

void length()

{

    char sen[100];

    printf("enter a sentence\n");

    fgets(sen, sizeof(sen), stdin);

    int count=0;

    int x=0;

    char ch=sen[x];

    while(ch!='\0' && sen[x] != '\n')

    {

        count++;

        x++;

    }

}
```

```

    printf("the length of the sentence is %d\n",count);
}
void print()
{
    char sen[100];
    printf("enter a sentence\n");
    fgets(sen,sizeof(sen),stdin);
    while (getchar() != '\n');
    int length=strlen(sen);
    printf("the characters of the string printed in each line\n");
    for(int i=0;i<length;i++)
    {
        printf("%c\n",sen[i]);
    }
}
void vowcon()
{
    char sen[100];
    printf("enter a sentence\n");
    fgets(sen,sizeof(sen),stdin);
    while (getchar() != '\n');
    int length=strlen(sen);
    int vow=0;
    int con=0;
    for(int i=0;i<length;i++)
    {
        sen[i]=tolower(sen[i]);
    }
}

```

```

for(int i=0;i<length;i++)
{
    if (sen[i] == 'a' || sen[i] == 'e' ||
        sen[i] == 'i' || sen[i] == 'o' ||
        sen[i] == 'u')
        vow++;
    else if(sen[i]!=' ' && sen[i]!='\n')
        con++;
}

printf("the no of vowels in the string are %d and consonent are%d",vow,con);
}

int main()
{
    int ch;

    printf("choice 1 is to find the length of a string(not using a in built function)\n");
    printf("choice 2 is to print each character of a string in a new line\n");
    printf("choice 3 is to Count vowels and consonants in a string.\n");
    printf("exit is choice 4\n");
    printf("enter your choice\n");
    scanf("%d",&ch);
    while (getchar() != '\n');
    switch (ch)
    {
        case 1:
        {
            length();
            break;
        }
    }
}

```

```
    case 2:
    {
        print();
        break;
    }
    case 3:
    {
        vowcon();
        return 0;
    }
    case 4:
    {
        printf("exit\n");
        return 0;
    }
    default:
    {
        printf("invalid choice\n");
        break;
    }
}
return 0;
}
```